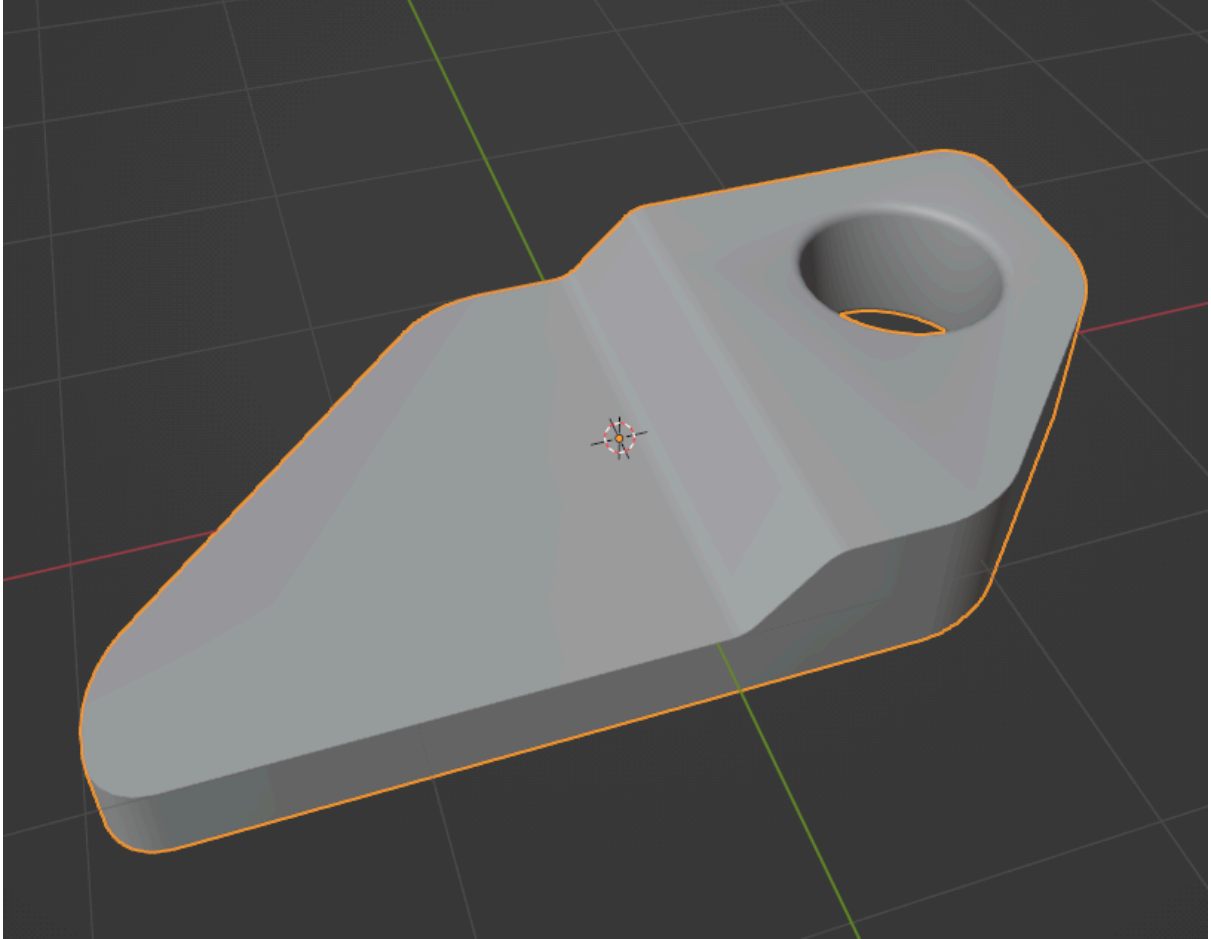
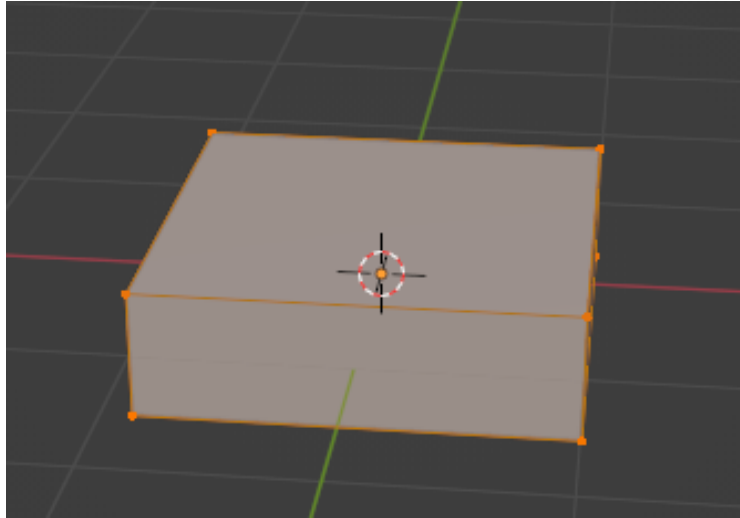


Hard Surface Model

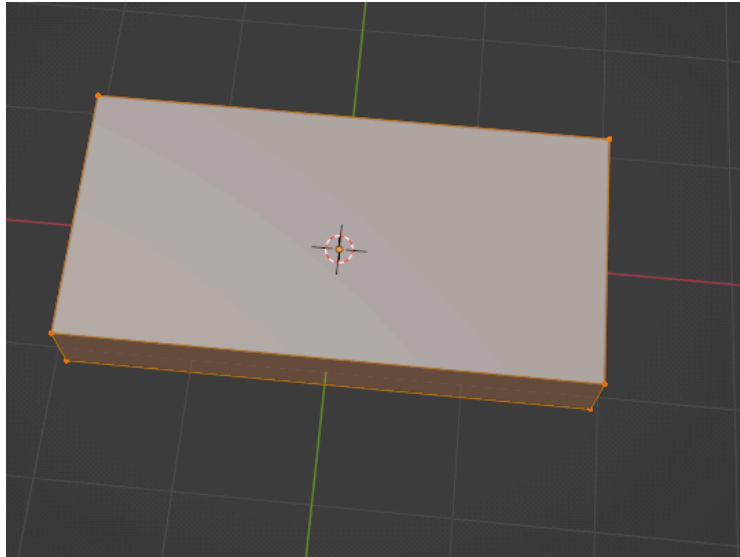
Important concepts covered: boolean modifier, bevel



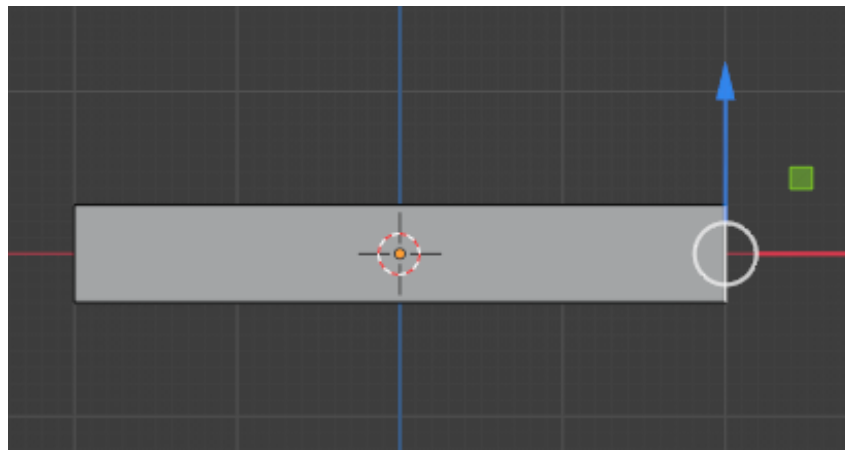
First, select the default cube and enter edit mode. Scale it down along the Z axis (S+Z) by 0.3:

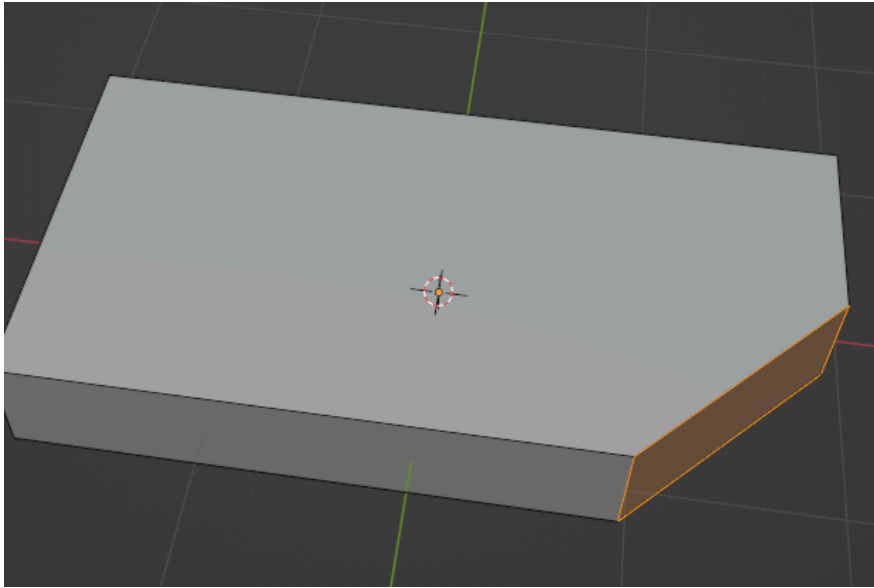


Next, scale it along X axis (S+X) by 2:

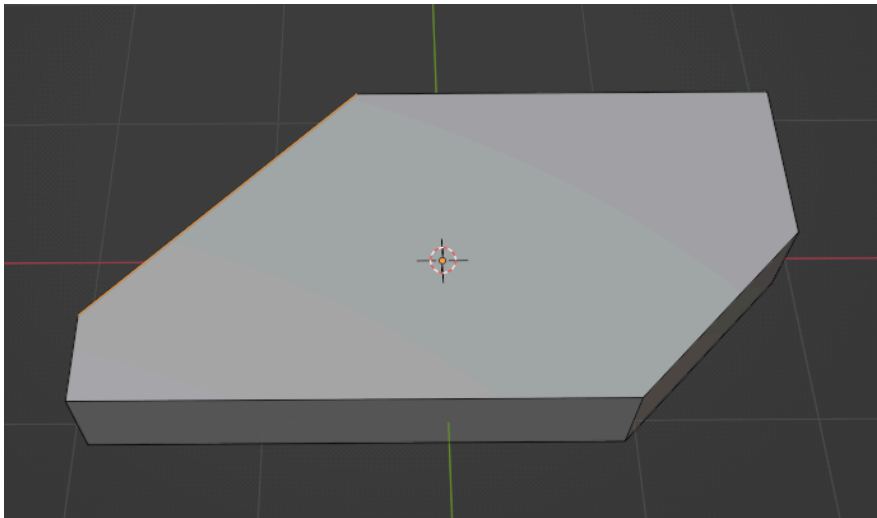


Now, go to the front view (NUMPAD 1), change to edge selection mode and select the right vertical edge and bevel it (CTRL + B) by 1:





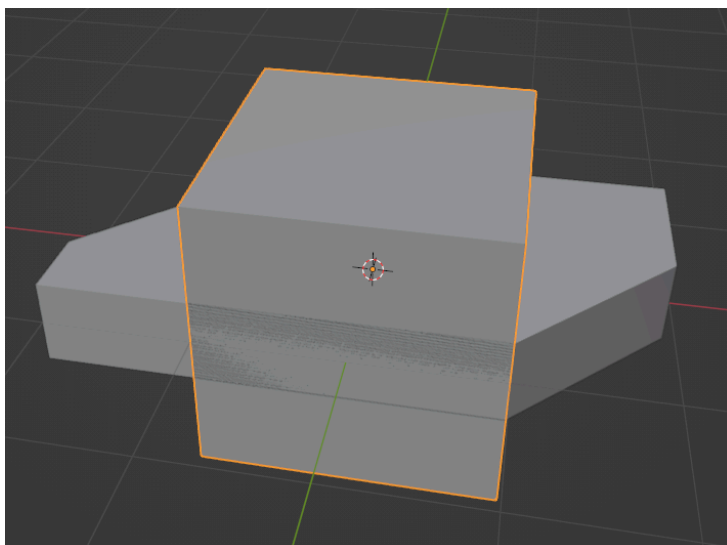
Next, select the opposite vertical edge and bevel (CTRL+B) it by 1.5:



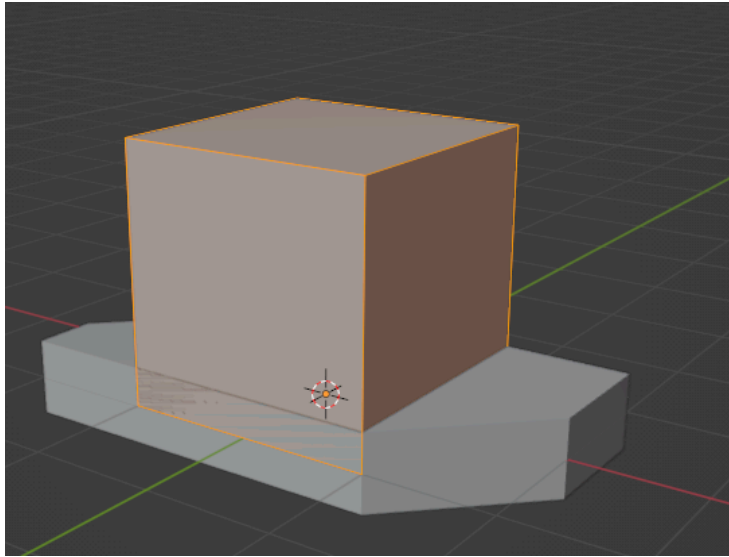
Now we have our **main piece**. Let's start now modeling the **details**:

Make sure that the 3D cursor is at the center (you can center it by pressing SHIFT + C).

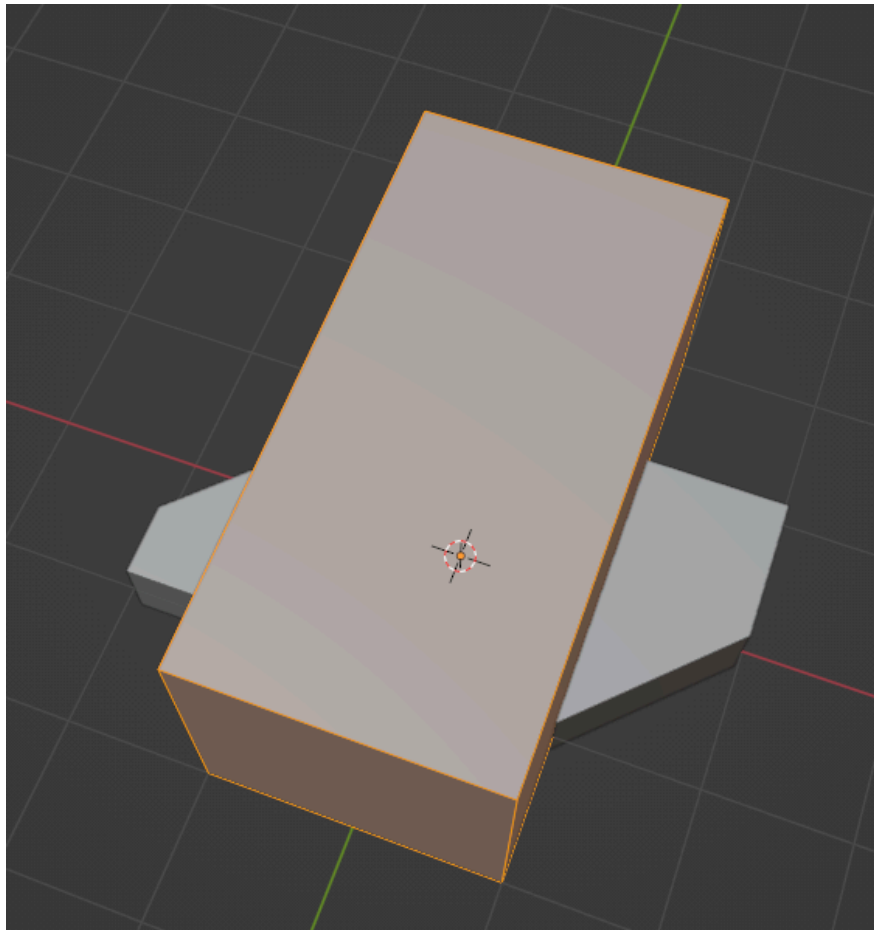
Go back to object mode and add a new cube (SHIFT+A → MESH → Cube):



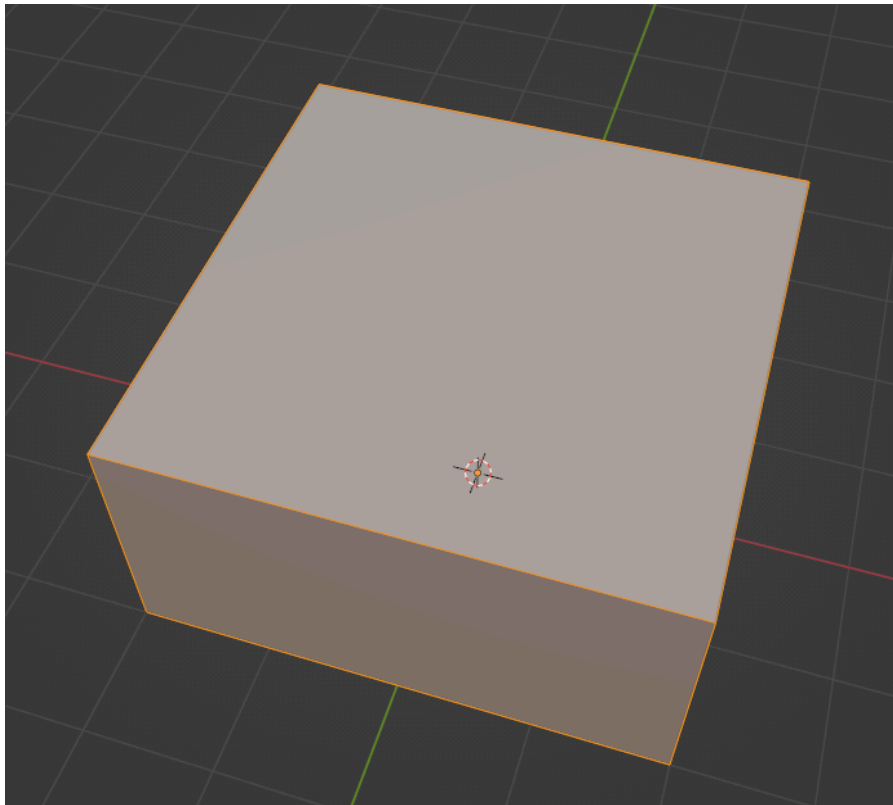
Enter edit mode and move it along Z axis (G+Z) by 1:



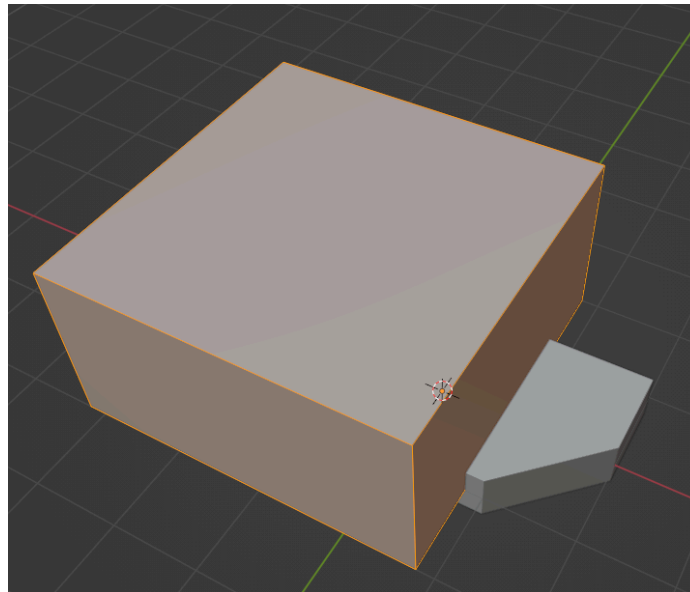
Scale it along Y axis (S+Y) by 2:



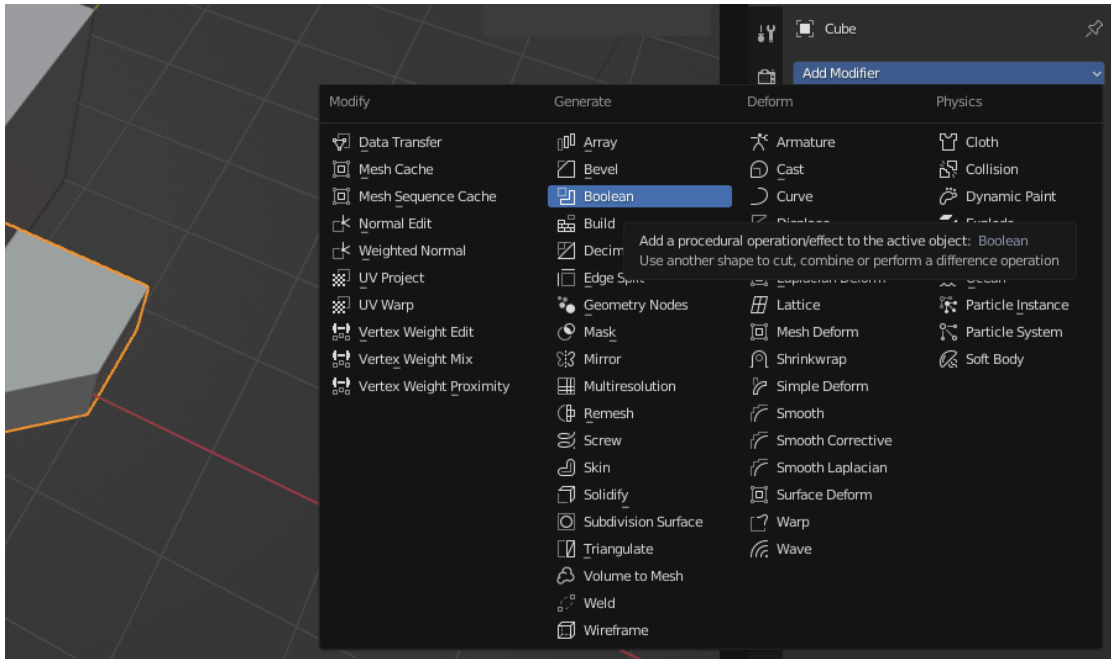
Do the same along X axis (scale by 2) $\rightarrow$ (S+X):



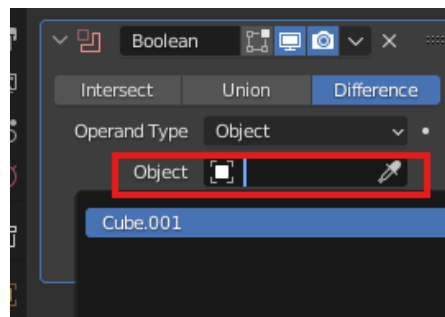
Now move it along the X axis (G+X) by -1.2:



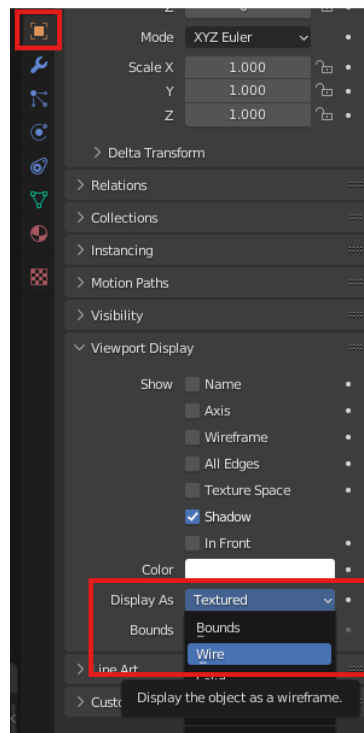
Now go back to object mode, select the **main piece** and add a boolean modifier:



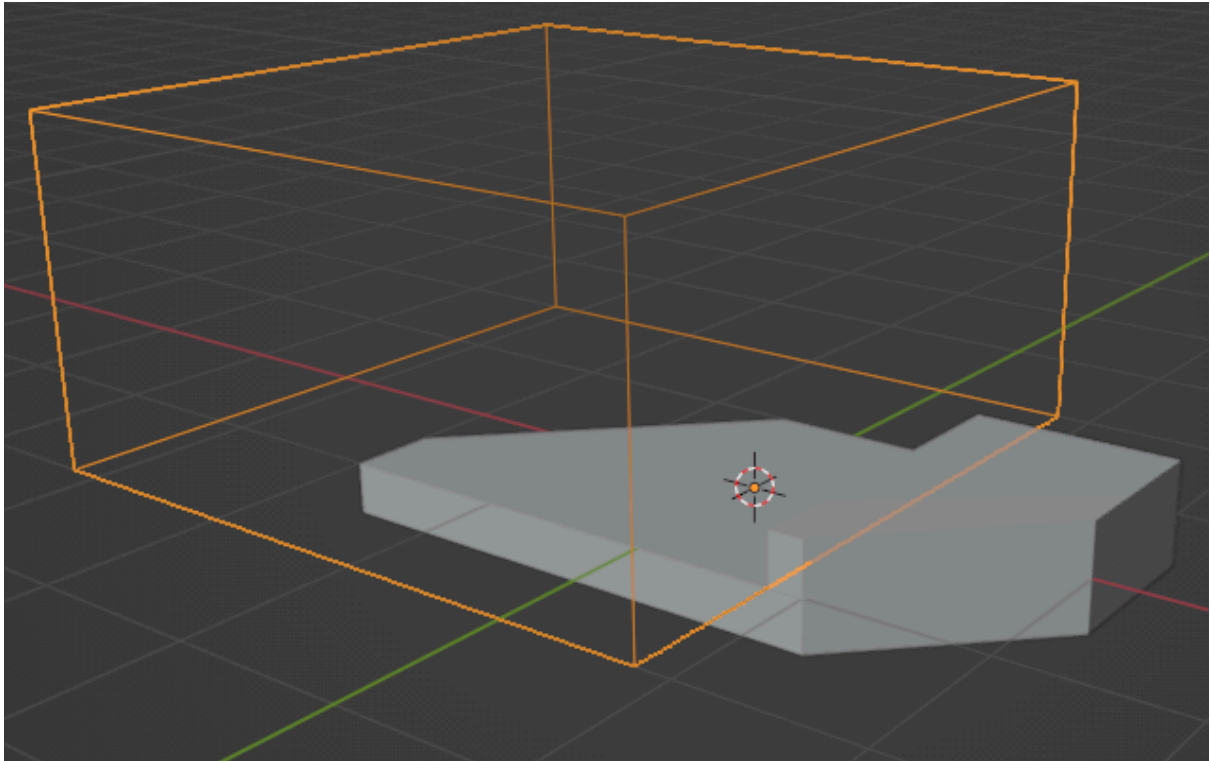
Under the “Object” property select “Cube.001”:



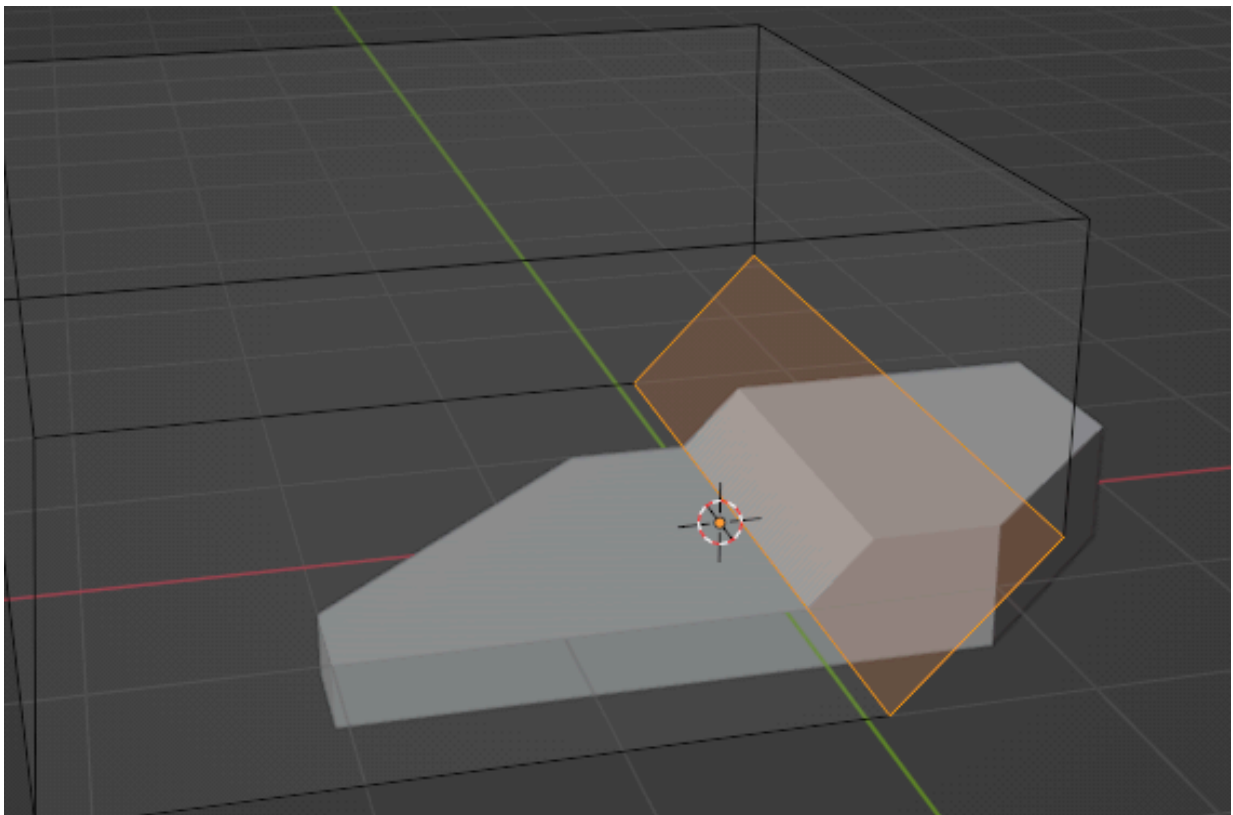
Now select the **details** object (Cube.001) again and go to “Object Properties” → “Viewport Display” → “Display As” and select “Wire”:



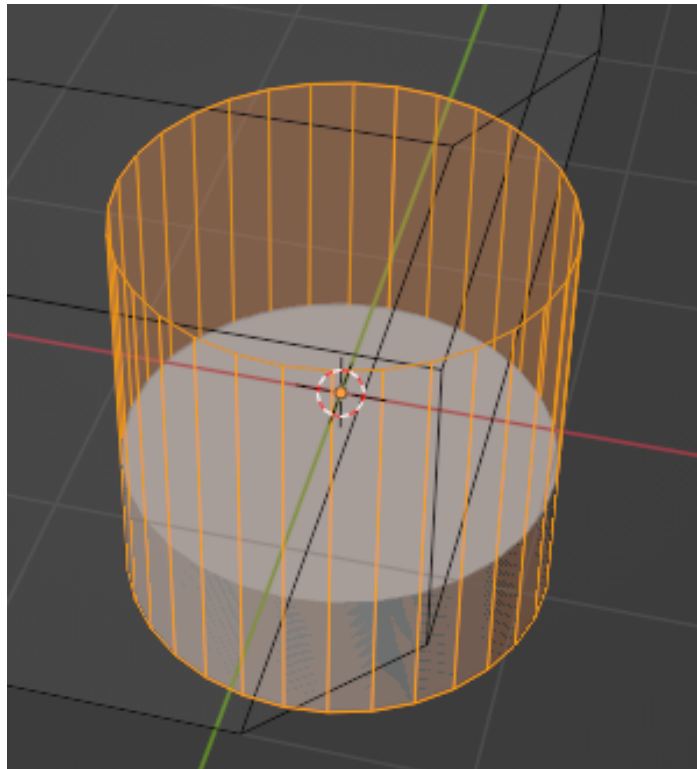
Now the cube should be displayed only as wire:



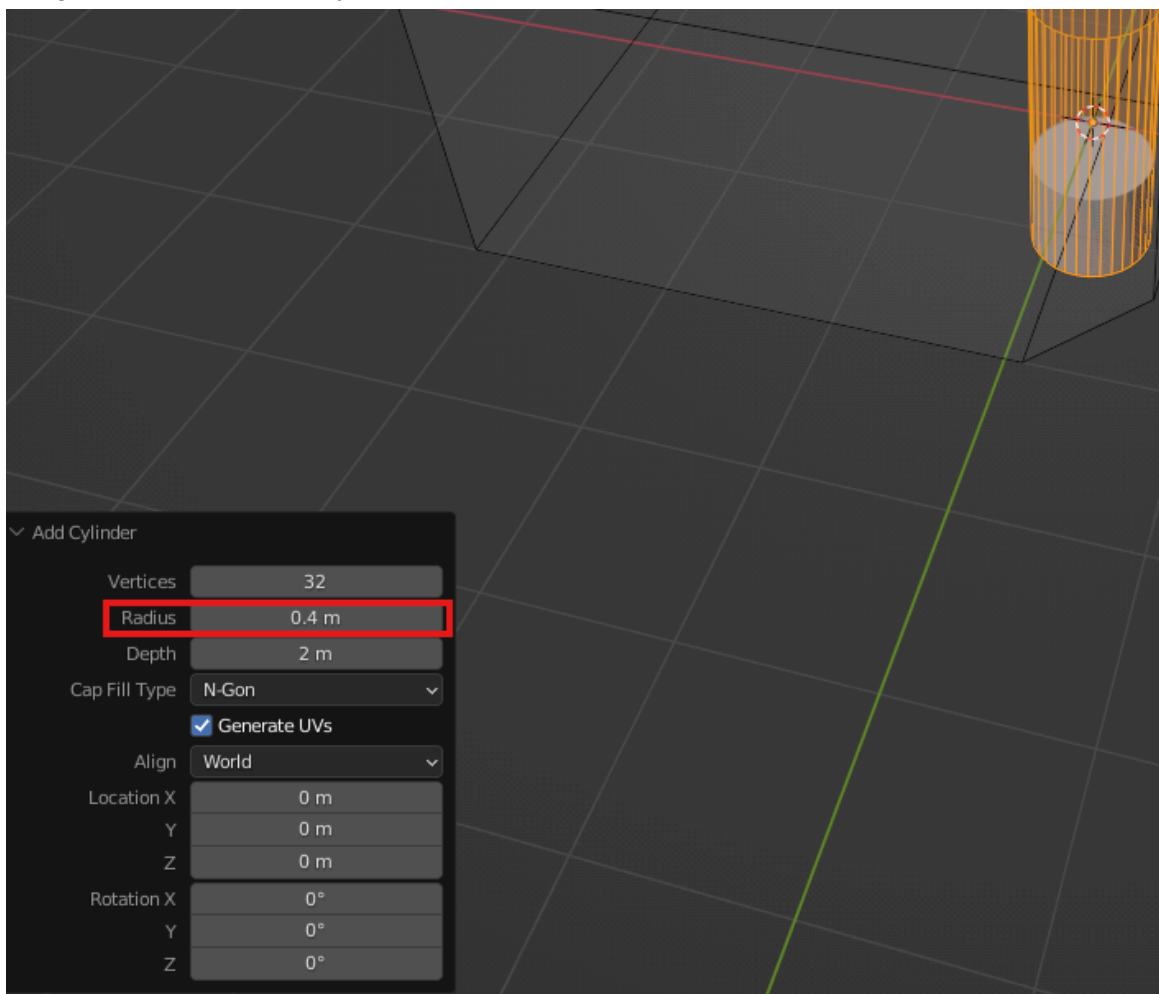
Next, with the cube still selected, enter edit mode, select the edge that is cutting the other object and bevel (CTRL+B) it by 0.7:



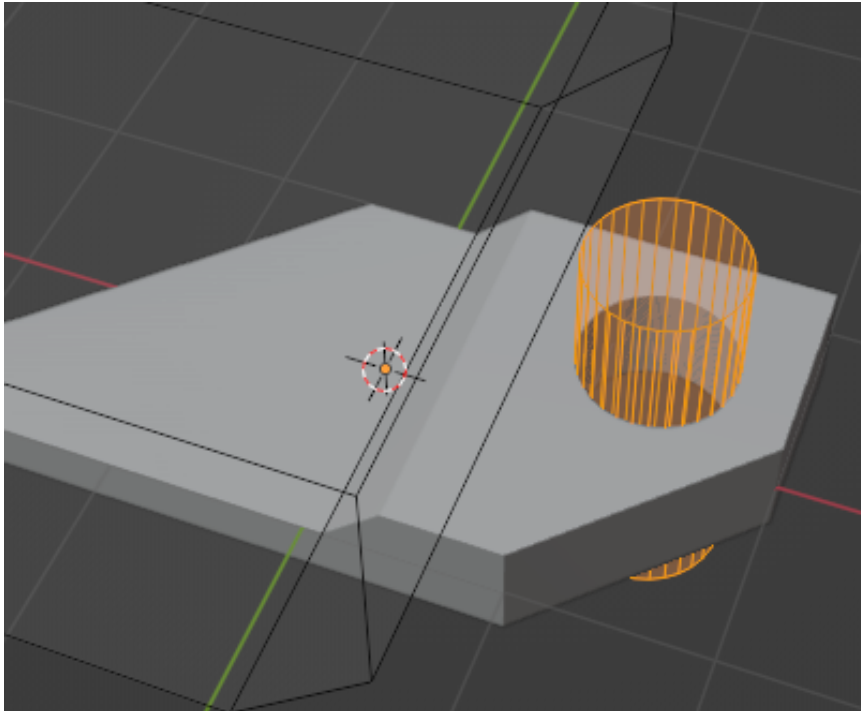
Now, make sure that the 3D cursor is at the center (you can center it by pressing SHIFT + C) and add a cylinder still in the edit mode of the object (SHIFT+A → MESH → Cylinder):



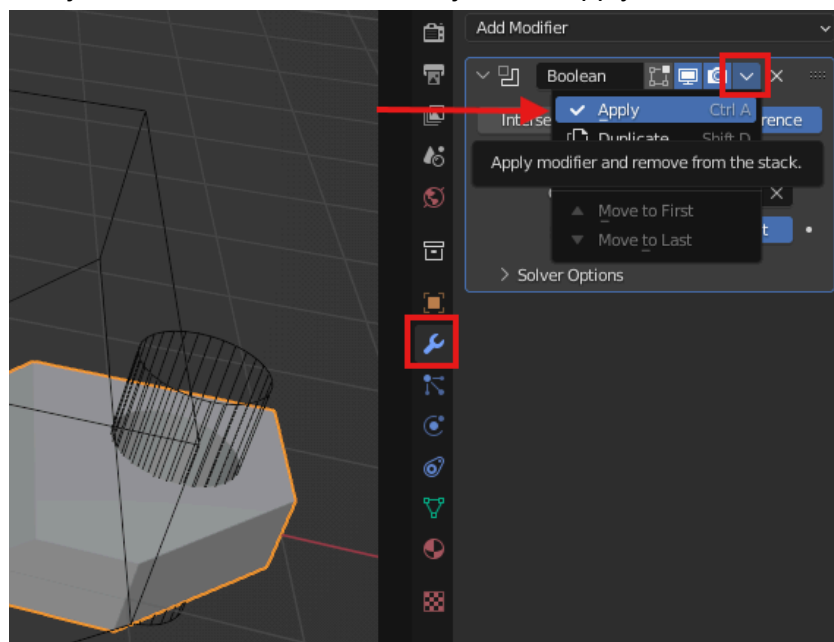
Change the Radius property to be 0.4:



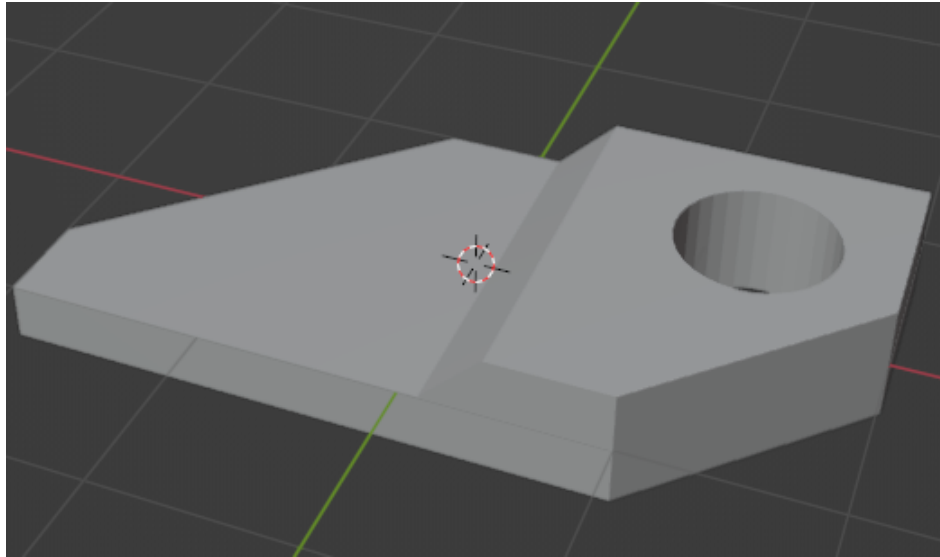
Move it along the x axis (G+X) by 1.3 and then along the Y axis (G+Y) by 0.3:



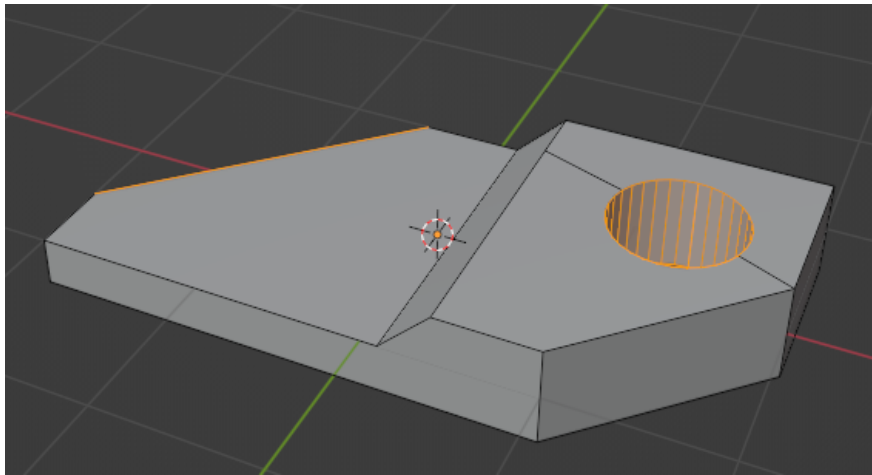
Now go back to object mode, select the other object and apply the modifier.



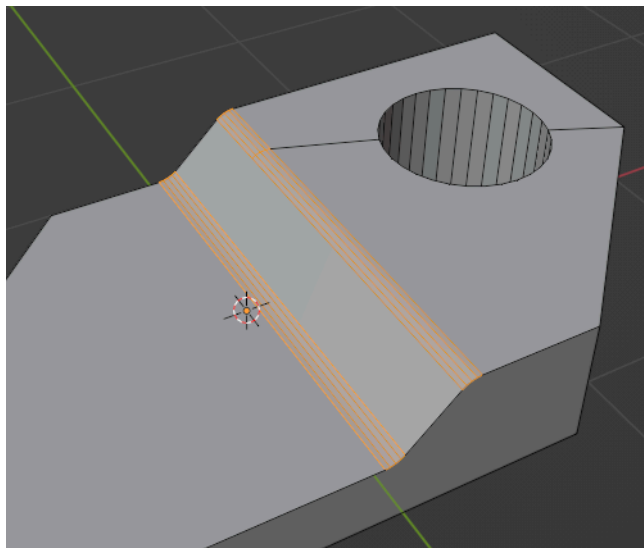
Now select the other object and press “H” to hide it from the viewport:



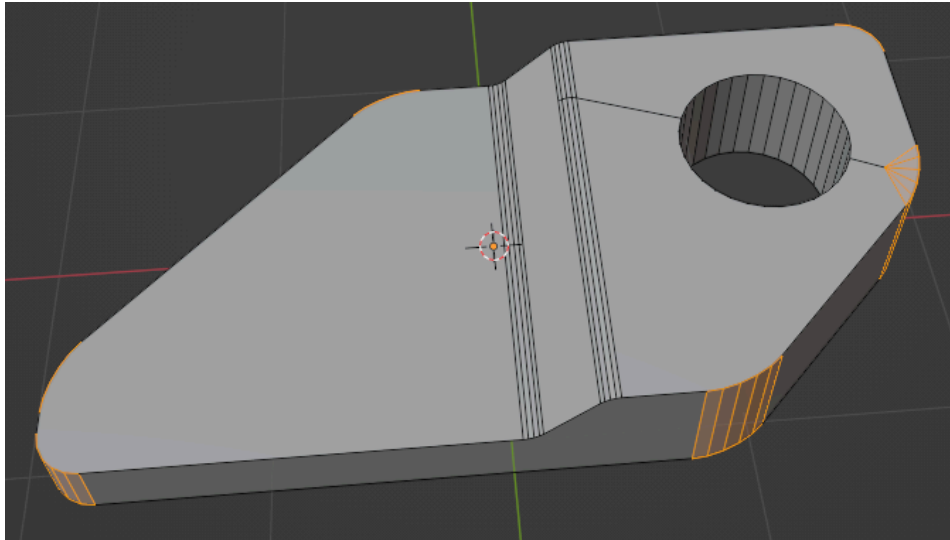
Now let's add some details to this piece.
First, select it and enter edit mode:



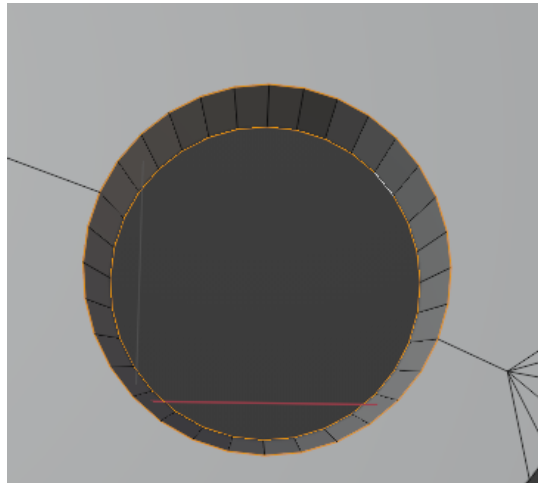
Now, select the outer horizontal edges of the “ramp” and bevel (CTRL+B) them by 0.05 and 4 segments (use mouse wheel to increase the segment number):



Now, let's also add a bevel (CTRL+B) to the outer vertical edges by 0.2 and using 5 segments:

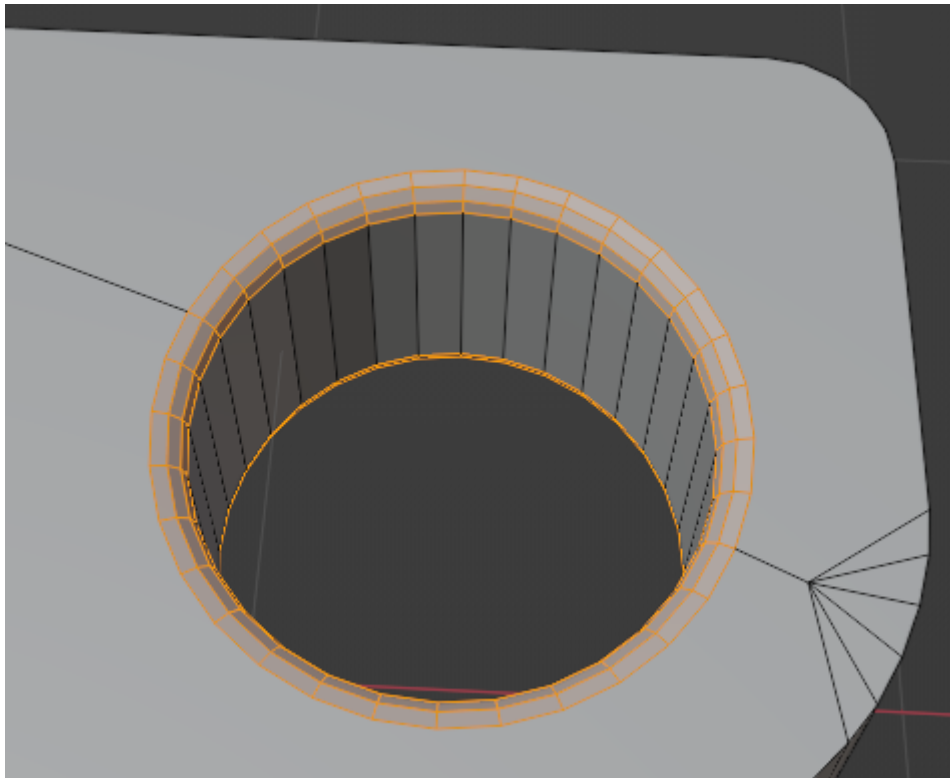


Finally, select the edge loops both of the top and bottom of the cylinder (to select an edge loop, hold ALT and click one of the edges. To select multiple loops, hold SHIFT + ALT and click one of the edges again):

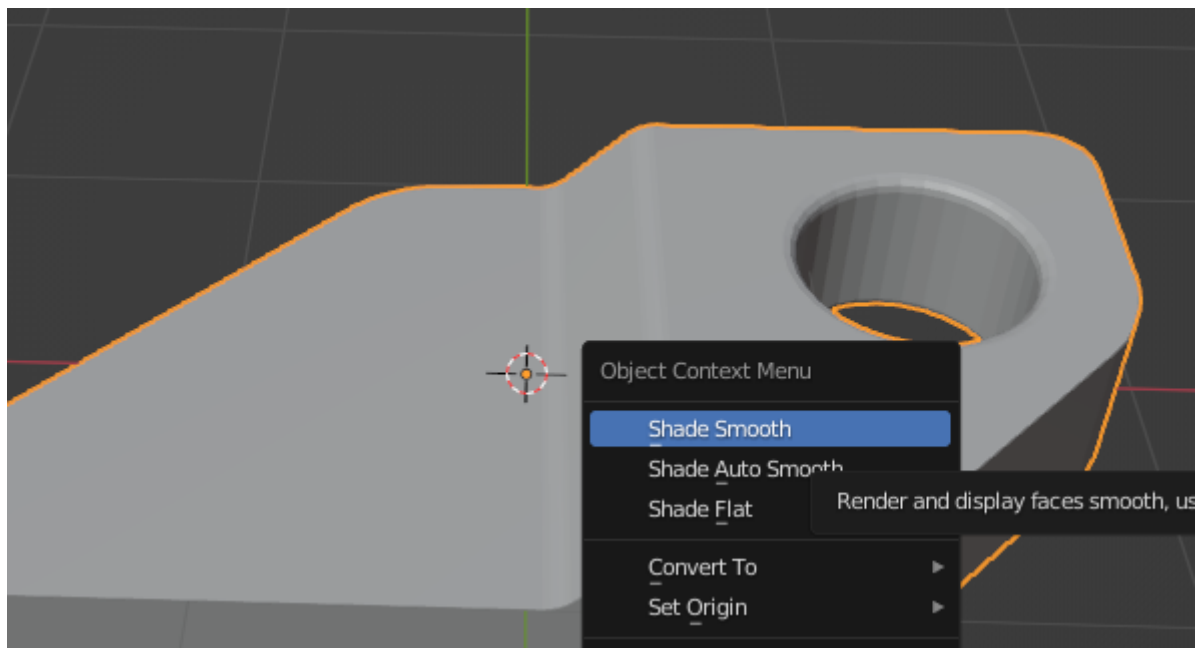


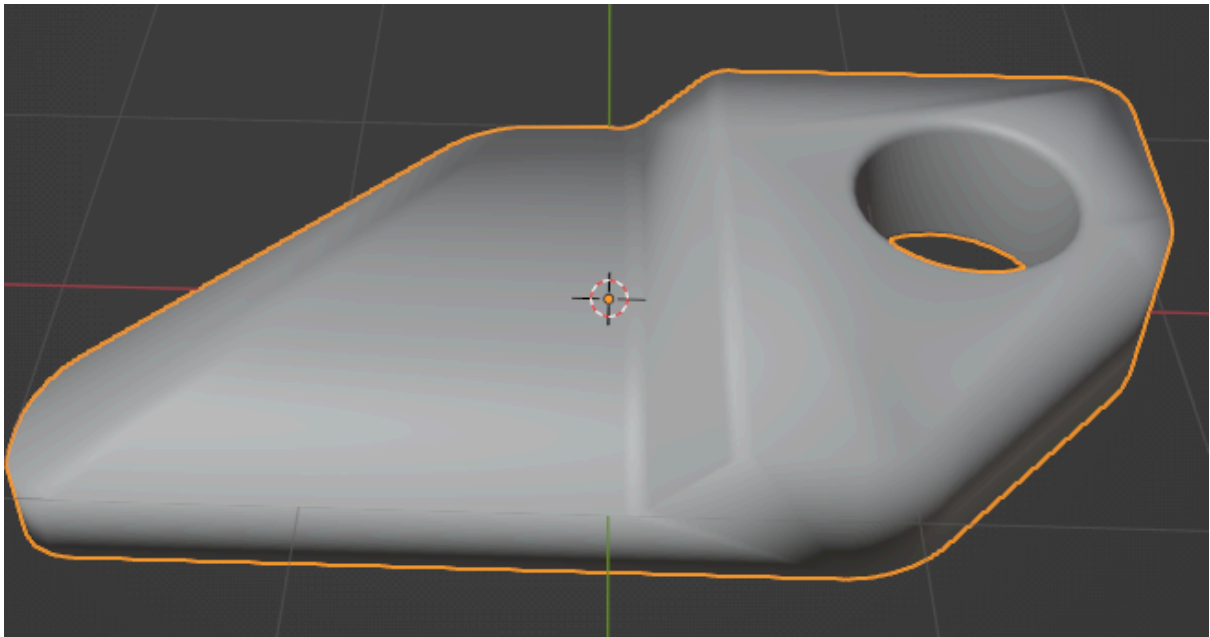
Don't forget to select all the edges of the circle. Since there is a cut in the mesh, if you ALT + Click it won't select the whole circle at once, you will have to repeat the selection of the edge loop until you have all the edges selected like in the previous image.

Now, add a bevel (CTRL+B) to the selection by 0.05 and 3 segments:

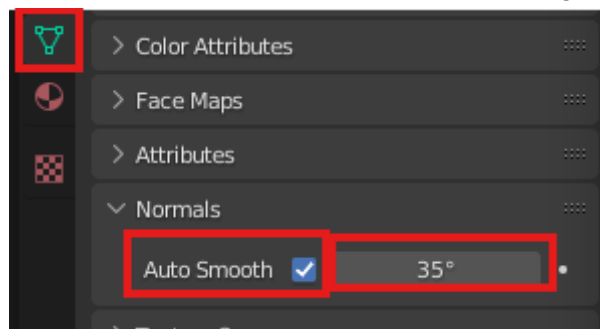


Now, go back to Object mode and right click on the viewport and select “Shade Smooth”:





As you can see, the mesh looks “weird”. That’s because it is smoothing all the vertices. We have to enable “Auto Smooth” to solve this issue. In order to do so, go to “Object Data Properties” → “Normals” and check “Auto Smooth” and set the angle to 35°:



Now you have the final piece!

